

Mathilde André, PhD - population genomicist

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I am a population genomicist with a strong **multidisciplinary** background that integrates genomics, **bioinformatics** and anthropology. Throughout my academic path, I have been committed to contextualising genomic results within the historical and cultural backgrounds of the communities studied. From 2019 to 2023, I pursued a PhD at the Institute of Genomics of the University of Tartu (Estonia) as part of the Papuan Past Project, an international and multidisciplinary collaboration combining archaeological and genomic approaches to explore human population settlement and adaptation in Papua New Guinea (PNG). My research focused on identifying **phenotypic adaptation and positive selection signatures** in two distinct populations: PNG highlanders from the Mt. Wilhelm region and lowlanders from Daru Island. I then developed a strong interest in **transcriptional variation** during my first postdoctoral position at the Computational Biology Unit of the University of Turin (Italy), where I created **prediction models** for genetically regulated gene expression (**GReX**) and alternative splicing in ancient European populations. Since January 2025, I have been a postdoctoral researcher in the Human Evolutionary Genetics Unit at the Institut Pasteur in Paris (France), where my work focuses on detecting **selective sweeps** and investigating **polygenic selection** in French Polynesians.

Work experience:

2025 – Now France	Postdoctoral researcher at the Human Evolutionary Genetics Unit, Institut Pasteur – from the 13/01/2025 Project: Exploring signals of polygenic selection in French Polynesian populations <u>Supervisors:</u> Pr L. Quintana-Murci, Dr E. Patin
2023-2024 Italy	Postdoctoral researcher at the department. of Neurosciences “Rita Levi Montalcini”, University of Turin – from the 01/09/2023 to the 30/11/2024 Project: Retrieving gene regulatory profiles from high-quality ancient genomes <u>Supervisors:</u> Pr P. Provero, Dr D. Marnetto

Education:

2019-2024 Estonia	PhD in Gene Technology, University of Tartu, Faculty of Science and Technology, Institute of Genomics, Chair of Evolutionary Biology – defended on 12/09/2023 Thesis: New Guinea, a hotspot for Human evolution: settlement history and adaptation in northern Sahul <u>Supervisors:</u> Dr M. Mondal, Dr F.-X. Ricaut
2017-2019 France	MSc, Evolutionary Anthropology (“Master Humanités Médicales”), University of Aix-Marseille Thesis: Facial skin ageing genetics <u>Supervisors:</u> Pr A. Ruiz-Linares, Dr P. Faux
2013-2017 Belgium	BSc, Major in Biological Sciences with Minor in History of Art and Archaeology, Catholic University of Louvain

Teaching:

2020-2021	Teaching assistant, University of Tartu, Bachelor in Science and Technology, Chair of Evolutionary Biology, Evolution and the Natural World course (12 hours)
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Publications:

2025	M André , F Montinaro, E Zapparoli, P Provero, D Marnetto, <i>Imputed transcriptional patterns in ancient genomes reveal molecular targets of selection</i> , biorxiv, 2025 Nov, https://doi.org/10.1101/2025.11.21.689732 - Preprint
2025	M Mondal, M André , A K Pathak, N Brucato, F-X Ricaut, M Metspalu, A Eriksson, <i>Resolving out of Africa event for Papua New Guinean population using neural</i>

- network*, Nat Com, 2025 Jun, <https://doi.org/10.1038/s41467-025-61661-w>
- 2025** D Fusco, C Marinelli, **M André**, L Troiani, M Noè, F Pizzagalli, D Marnetto, P Provero, *Exploring the molecular basis of the genetic correlation between body mass index and brain morphological traits*, PLoS Genet, 2025 Apr, <https://doi.org/10.1101/2024.08.09.24311778>
- 2024** D Yermakovich, **M André**, N Brucato, J Kariwiga, M Leavesley, V Pankratov, M Mondal, F-X Ricaut, M Dannemann, *Denisovan admixture facilitated environmental adaptation in Papua New Guinean populations*, PNAS, 2024 Jun, 121(6), <https://doi.org/10.1073/pnas.2405889121>
- 2024** **M André**, N Brucato, G Hudjasov, V Pankratov, D Yermakovich, R Kreevan, J Kariwiga, J Muke, [...] , M Mondal, F-X Ricaut, *Positive selection in the genomes of two Papua New Guinean populations at distinct altitude levels*, NatCom, 2024 Apr, <https://doi.org/10.1038/s41467-024-47735-1>
- 2022** N Brucato, **M André**, G Hudjasov, M Mondal, M P Cox, M Leavesley, F-X Ricaut, *Chronology of natural selection in Oceanian genomes*, IScience, 2022 Jun, 25(7), <https://doi.org/10.1016/j.isci.2022.104583>
- 2021** Y Chen*, **M André***, K Adhikari, M Blin, B Bonfante, J Mendoza-Revilla, M Fuentes-Guajardo, S Palmal, [...] , P Faux , A Ruiz-Linares; *A genome-wide association study identifies novel gene associations with facial skin wrinkling and mole count in Latin Americans*; British Journal of Dermatology; 2021 Nov; 18(5):988-998; doi:10.1111/bjd.20436
- 2021** N Brucato, **M André**, R Tsang, L Saag, J Kariwiga, K Sesuki, T Beni, W Pomat, [...], F-X Ricaut; *Papua New Guinean Genomes Reveal the Complex Settlement of North Sahul*; Molecular Biology and Evolution; 2021 Oct; 38(11):5107-5121; doi:10.1093/molbev/msab238
- 2021** **M André**, N Brucato, S Plutniak, J Kariwiga, J Muke, A Morez, M Leavesley, M Mondal, F-X Ricaut; *Phenotypic differences between highlanders and lowlanders in Papua New Guinea*; PLOS ONE; 2021 Jul; 16(7): e0253921; doi:10.1371/journal.pone.0253921
- 2020** N Pedro, N Brucato, V Fernandes, **M André**, L Saag, W Pomat, C Besse, A Boland, [...], F-X Ricaut; *Papuan mitochondrial genomes and the settlement of Sahul*; Journal of Human Genetics, 2020 Jun; 65:875:887; doi: 10.1038/s10038-020-0781-3

International Conferences:

- 2023** The Society for Molecular Biology and Evolution (SMBE) meeting 2023, Ferrara, Italy
Poster: Genomic Signatures of Local Adaptation to Extreme Environments in Papua New Guinean Highlanders and Lowlanders
- 2023** The Biology of Genomes, Cold Spring Laboratory (CSHL), USA
Poster: Positive selection in the genomes of two Papua New Guinean populations at distinct altitude levels
- 2022** Congress of the Indo-Pacific Prehistory Association (IPPA), Chiang Mai, Thailand
Oral presentation: Phenotype Adaptation and Positive Selection to Altitude in Papua New Guinean Highlanders
- 2022** EMBO | EMBL Symposium: Reconstructing the human past: using ancient and modern genomics, Heidelberg, Germany
Flask talk and poster: Papuan New Guineans blood composition is affected by positive selection
- 2021** Australian Archeological Association conference, online
Oral presentation: Papua New Guineans Show Signs of Biological Adaptations to Altitude
- 2021** Wellcome Connecting Science conference – Human Evolution – From Fossils to Ancient and Modern Genomes, online
Poster: Natural Selection and phenotypic associations in Papuan Highlanders

- 2021** ASHG virtual meeting 2021, online
Poster: Positive selection to altitude and phenotypic associations in Papua New Guinea highlanders
- 2021** 1st Virtual Conference for Women Archaeologists and Paleontologists, online
Oral presentation: Papua New Guineans show unique phenotypic traits at altitude

Awards:

- 2023** Mobility grant of the Doctoral School of Biomedicine and Biotechnology.
 Funding of the attendance to “The Biology of Genomes” 2023 conference at the CSHL
- 2022** Dora Pluss Scholarship
 Funding to study polygenic selection in Papuan Highlanders at the Department of Genomes and Genetics at the Institut Pasteur in Paris, France, under the supervision of Dr Etienne Patin.

Public outreach

- 2024** Press article: *Denisovan DNA may help modern humans adapt to different environments*, New Scientist, <https://www.newscientist.com/article/2438941-denisovan-dna-may-help-modern-humans-adapt-to-different-environments/> (12/07/2024)
- 2024** Radio interview: *Papua New Guinean DNA offers insight into disease resilience*, Pacific Beat, ABC radio, <https://www.abc.net.au/pacific/programs/pacificbeat/png-superman-dna/103936504> (04/06/2024)
- 2024** Press article: Genetic adaptations have impacted the blood compositions of two populations from Papua New Guinea, Research in Estonia, <https://researchinestonia.eu/2024/05/16/genetic-changes-and-blood> (30/04/2024)
- 2021** Seminar to undergraduate and the general audience: *Étude phénotypique en Papouasie Nouvelle-Guinée : adaptation biologique à l'altitude chez les Papous des Hautes-Terres ?*, Midis des sciences naturelles, Université du Québec à Rimouski, Canada (18/11/2021)
- 2021** Radio interview: *Les Papous comme modèle d'adaptation humaine à l'altitude*, Les années lumière, Radio Canada, <https://ici.radiocanada.ca/ohdio/premiere/emissions/les-annees-lumiere/episodes/558960/rattrapagedu-dimanche-8-aout-2021> (08/08/2021)
- 2021** Press article: *Why Papua New Guinea's Highlanders Differ Physically From Those Living Near Sea Level*, Smithsonian Magazine, <https://www.smithsonianmag.com/science-nature/papua-new-guineas-highlandersdiffer-physically-individuals-living-near-sea-level-180978222/> (21/07/2021)